Hycient Igweze

AI / ML Engineer · Full-Stack
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PROFILE SUMMARY

- ML and full-stack engineer with ~a decade of experience shipping production systems, now focused on putting models into products: recommendation systems, LLM agents and the data pipelines behind them.
- At Whiteshield, wrote the skills and course recommendation model for UAE Skills, a national platform for the UAE government, and helped build the EduHub admissions portal.
- Design LLM systems where the model proposes and deterministic code decides, and evaluate models with leakage-safe splits and honest metrics (PR-AUC, grouped splits, bootstrap tests).
- Deep production background in fintech and payments: PayS ($30M+ processed monthly) and a multi-currency crypto exchange.

CORE COMPETENCIES

Recommendation Systems · LLM Agents & Guardrails · ML Pipelines & Model Evaluation · Data Engineering & De-identification · AI/LLM Product Engineering · System Architecture · API Design · Stakeholder Management

TOOLS AND TECHNOLOGIES

- **AI & ML:** Recommendation models, LLM agents (LangGraph, Claude API, ChatGPT API, Zod structured outputs), embeddings, scikit-learn, TensorFlow/Keras, model evaluation, data pipelines
- **Languages:** TypeScript, JavaScript, C#, Python, Dart
- **Frontend & Mobile:** React, Next.js, React Native, Flutter, Tailwind CSS
- **Backend & Data:** Node.js, NestJS, .NET, PostgreSQL, MongoDB, Prisma, GraphQL
- **Cloud & Delivery:** AWS, Azure, Railway, Cloudflare, CI/CD, TDD

SELECTED ML & AI PROJECTS

Handshake Agent (LLM Payments Agent) *NestJS, Next.js, LangGraph, Claude, Prisma, PostgreSQL*

- Chat-native crypto and payments agent where the LLM cannot move money: its only output is a validated Zod intent with no destination field, and a deterministic engine re-derives every amount, re-runs KYC, velocity and sanctions checks at settlement, and executes only with a PIN and a single-use HMAC grant on a BigInt double-entry ledger. About 78 lines of LLM code against 3,000 of engine. Write-up | Code

Credit Scoring on African Loan Data *scikit-learn, TensorFlow/Keras*

- 68,000 loans with a 1.8% default rate; customer-grouped splits, history features built only from earlier loans, and PR-AUC instead of accuracy. A tuned Random Forest (PR-AUC 0.936, 92% recall at 77% precision) beat a wide & deep network, with the gap confirmed by a customer-level bootstrap. Article

Malaria Detection CNN *TensorFlow/Keras, MobileNetV2*

- Lightweight blood-smear classifier split by source image to prevent leakage: 95.6% sensitivity, 96.5% specificity and 0.991 ROC-AUC on 4,105 held-out cells. Article

Multimodal Identity & Recommendation Pipeline *Python*

- Built the execution layer that chains face recognition, voice verification and product recommendation models into one gated flow (team project). Code

Medical Imaging Data Pipeline (Blimdata) *Python, NestJS, Next.js*

- Pipeline that turns hospital imaging archives into de-identified, annotated datasets for AI teams: read-only PACS gateway, DICOM de-identification of 18 identifiers plus burned-in text, radiologist review of AI-proposed segmentations, and versioned datasets with provenance. Built with the Blimdata team through PlumAutomate. blimdata.tech

PROFESSIONAL EXPERIENCE

Software Engineer – Whiteshield Group | Remote **Jan. 2025 – Present**

- **UAE Skills** (MoHRE & MoHESR): wrote the skills and course recommendation model behind the student experience, and worked across the full platform, a web, iOS and Android service linking labour-market demand, graduate outcomes and enrolment.

- Build AI-powered national platforms for UAE government clients as part of a 14-person engineering team, using Next.js, TypeScript and .NET (C#).
- **EduHub** (MoHESR): part of the team that developed the national admissions and scholarship portal, with 74 institutions registered and 40,000+ students expected in the current phase; its AI-enhanced scholarship flow cut applications from 22 form fields to four.
- Engineered an AI-driven job listing and candidate-matching system, cutting job description drafting time by about 70%.
- Collaborate with policy and economics teams across 4 continents to turn strategy into production software that meets public-sector QA standards.

Software Engineer – Sahara Group | Remote **Oct. 2023 – Sept. 2025**

- Built core services for PayS, an energy and utility payments platform whose largest client is Ikeja Electric, processing $30M+ monthly for 500,000+ active users, including the receipt service behind every transaction.
- Led the PayS Store multi-vendor marketplace (1,000+ vendors) and the Flutter mobile app, and built an in-house CDN on AWS S3 shared across products.

Lead Mobile App Engineer – Cloud P2P | Remote **Aug. 2022 – Oct. 2023**

- Led a team of 3 mobile engineers to ship 2 major versions of a peer-to-peer crypto trading platform, reaching 10,000+ users within 2 months, with multi-currency and cross-chain trading, escrowed trades, ratings and dispute resolution.
- Shaped the app architecture and business logic, and raised code quality through rigorous code reviews and a structured release process.

Lead Frontend Engineer – Pluralcode | Remote **May 2022 – Jul. 2023**

- Developed and managed a Learning Management System supporting 10,000+ monthly users, and maintained the ACHAN (Airport Car-Hire Association of Nigeria) booking system for 3,000+ customers and providers.
- Developed an optician's e-commerce API in Node.js integrated with a React frontend, and mentored cohorts of new engineers in JavaScript and React.

Freelance & Contract Engineer – Remote **2018 – 2022**

- Node.js commerce APIs (Wetfire), a React e-commerce platform (Adehot), React Native fintech apps (Vision Savings, Tranxnow), DeFi lending flows (Vilo), a reusable component library (Groomlog) and iOS telehealth apps for patients and doctors (myhealthXP).

VENTURES

- **PlumAutomate** – Founder, software and automation studio. Work includes the Zentry ticket marketplace (web, iOS and Android; 10,000+ users and 100,000+ tickets in its first year), a cross-border settlement API for Handshake, and Blimdata's medical-imaging pipeline (PACS gateway, DICOM de-identification, radiologist annotation). plumautomate.com
- **Moving Man Logistics** – Co-founder & CEO, on-demand delivery and rides in Kigali, Rwanda.
- **EcoRed** – Co-founder, SON-certified low-carbon laterite cement and bricks for affordable housing in Nigeria.

EDUCATION & SPEAKING

- **African Leadership University** – BSc, Machine Learning & Computer Science
- **Guest Speaker** – University of Greenwich International College (2026): “What it means to be an engineer when AI can write the code”, delivered to the new international engineering intake.